

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Instrumentation and Control Engineering) SEMESTER - 3 Winter 2025 ( Regular )**

**Course :Bachelor of Technology (Instrumentation and Control Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 3**

**Subject Code & Name: 25AF1464PC302 - SENSORS AND TRANSDUCERS**

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**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
  2. Question No. 1 will be compulsory and include objective-type questions.
  3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
  4. Use of non-programmable scientific calculators is allowed.
  5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12**
- 1 Which of the following is a dynamic characteristic of a transducer?
    - a) Accuracy
    - b) Sensitivity
    - c) Response time
    - d) Linearity
  - 2 Which transducer is commonly used for torque measurement?
    - a) Strain gauge
    - b) Thermistor
    - c) Thermocouple
    - d) Tachometer
  - 3 Which material is used in piezoelectric transducers?
    - a) Silicon
    - b) Quartz
    - c) Copper
    - d) Germanium
  - 4 Thermistors are
    - a) Negative temperature coefficient devices
    - b) Positive temperature coefficient devices
    - c) Both NTC and PTC
    - d) Voltage-dependent resistors
  - 5 Toothed rotor tachometer works on which principle?
    - a) Thermal
    - b) Magnetic induction
    - c) Piezoelectric
    - d) Photoelectric
  - 6 Magnetic flowmeters require the fluid to be
    - a) Non-conductive
    - b) Highly viscous
    - c) Conductive
    - d) Gaseous
  - 7 Viscosity measured using a falling ball type viscometer depends on:
    - a) Magnetic field
    - b) Thermal conduction
    - c) Terminal velocity of ball
    - d) Radiation

- 8 Which is variable head flow meter?  
a. Venturi meter  
b. Magnetic flow meter  
c. Turbine flow meter  
d. All
- 9 Dead weight tester is used for  
a) Pressure calibration  
b) Flow measurement  
c) Level measurement  
d) Temperature calibration
- 10 A float-type device is used for measuring  
a) Pressure  
b) Level  
c) Flow  
d) Speed
- 11 pH is defined as  
a) Concentration of salt  
b) Log of hydrogen ion activity  
c) Log of temperature  
d) Electrical conductivity
- 12 Conductivity is measured in  
a) Siemens/meter  
b) Pascal  
c) Joules  
d) Farad
- Q2. Solve the following.
- A) Classify strain gauges explain any one in detail 6
- B) List the selection criteria used while choosing a sensor for any measurement application 6
- Q3. Solve the following.
- A) Suggest suitable sensor used to measurement of angular speed 6
- B) With neat diagram explain construction and working of LVDT. 6
- Q4. Solve Any Two of the following.
- A) Explain construction and working of RTD. 6
- B) Explain construction and working of electromagnetic flow meter. 6
- C) Explain ultrasonic flow meter. 6
- Q5. Solve Any Two of the following.
- A) Explain construction and working of turbine flow meter 6
- B) Describe C tube type bourdon tube. 6
- C) Classify manometer and explain U tube manometer 6
- Q6. Solve Any Two of the following.
- A) Explain construction working of hygrometer 6
- B) Describe infrared moisture measurement 6
- C) Explain pH sensor 6

**\*\*\* End \*\*\***